

CS211 HW3 Sieving Prime

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Q 1

(10 points)

 <code>sieve0_1.o</code> ×	 <code>starter.py</code>	
<code>cs211-hw3-sieving-prime-pi1grim2019 > sh >  sieve0_1.o</code>		
<code>1 func=sieve0, pnum=32, count=455052511, time=11.915854</code>		
 <code>sieve0_2.o</code> ×	 <code>starter.py</code>	
<code>cs211-hw3-sieving-prime-pi1grim2019 > sh >  sieve0_2.o</code>		
<code>1 func=sieve0, pnum=64, count=455052511, time=8.362654</code>		
 <code>sieve0_3.o</code> ×	 <code>starter.py</code>	
<code>cs211-hw3-sieving-prime-pi1grim2019 > sh >  sieve0_3.o</code>		
<code>1 func=sieve0, pnum=96, count=455052511, time=5.455526</code>		
 <code>sieve0_4.o</code> ×	 <code>starter.py</code>	
<code>cs211-hw3-sieving-prime-pi1grim2019 > sh >  sieve0_4.o</code>		
<code>1 func=sieve0, pnum=128, count=455052511, time=4.068797</code>		
 <code>sieve0_5.o</code> ×	 <code>starter.py</code>	
<code>cs211-hw3-sieving-prime-pi1grim2019 > sh >  sieve0_5.o</code>		
<code>1 func=sieve0, pnum=160, count=455052511, time=3.172314</code>		

Q 2

(30 points)

 sieve1_1.o	 starter.py	
cs211-hw3-sieving-prime-pi1grim2019 > sh >  sieve1_1.o		
1 func=sieve1, pnum=32, count=455052511, time=5.620498		
 sieve1_2.o	 starter.py	
cs211-hw3-sieving-prime-pi1grim2019 > sh >  sieve1_2.o		
1 func=sieve1, pnum=64, count=455052511, time=3.094943		
 sieve1_3.o	 starter.py	
cs211-hw3-sieving-prime-pi1grim2019 > sh >  sieve1_3.o		
1 func=sieve1, pnum=96, count=455052511, time=2.124454		
 sieve1_4.o	 starter.py	
cs211-hw3-sieving-prime-pi1grim2019 > sh >  sieve1_4.o		
1 func=sieve1, pnum=128, count=455052511, time=1.608918		
 sieve1_5.o	 starter.py	
cs211-hw3-sieving-prime-pi1grim2019 > sh >  sieve1_5.o		
1 func=sieve1, pnum=160, count=455052511, time=1.586663		

Q 3

(30 points)

C sieve2.c 2, M $\equiv$ sieve2_1.o M $\times$

```
cs211-hw3-sieving-prime-pi1grim2019 > sh >  $\equiv$  sieve2_1.o
1 | func=sieve2, pnum=32, count=455052511, time=8.919830
```

C sieve2.c 2, M $\equiv$ sieve2_2.o M $\times$

```
cs211-hw3-sieving-prime-pi1grim2019 > sh >  $\equiv$  sieve2_2.o
1 | func=sieve2, pnum=64, count=455052511, time=4.472515
```

C sieve2.c 2, M $\equiv$ sieve2_3.o M $\times$

```
cs211-hw3-sieving-prime-pi1grim2019 > sh >  $\equiv$  sieve2_3.o
1 | func=sieve2, pnum=96, count=455052511, time=2.953022
```

C sieve2.c 2, M $\equiv$ sieve2_4.o M $\times$

```
cs211-hw3-sieving-prime-pi1grim2019 > sh >  $\equiv$  sieve2_4.o
1 | func=sieve2, pnum=128, count=455052511, time=2.193398
```

C sieve2.c 2, M $\equiv$ sieve2_5.o M $\times$

```
cs211-hw3-sieving-prime-pi1grim2019 > sh >  $\equiv$  sieve2_5.o
1 | func=sieve2, pnum=160, count=455052511, time=1.770180
```

Q 4

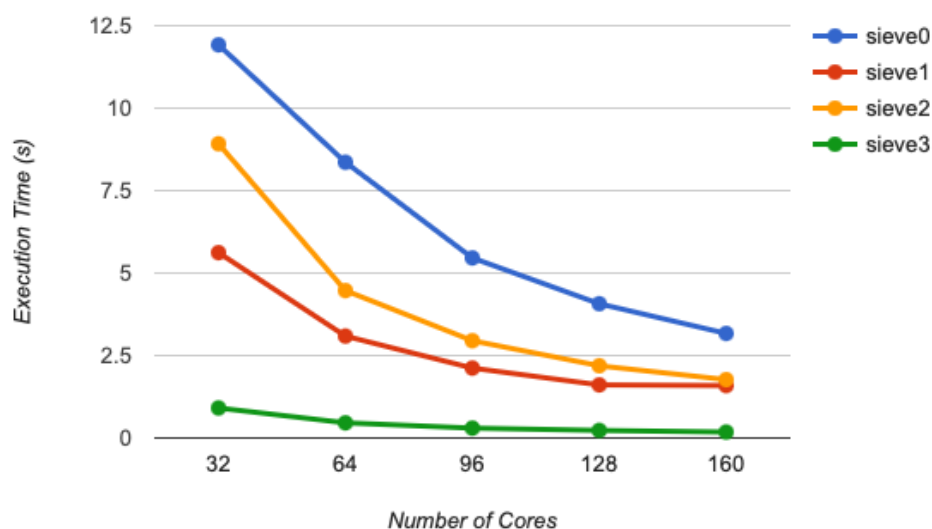
(30 points)

```

≡ sieve3_1.o × starter.py
cs211-hw3-sieving-prime-pi1grim2019 > sh > ≡ sieve3_1.o
1 func=sieve3, pnum=32, count=455052511, time=0.908336
≡ sieve3_2.o × starter.py
cs211-hw3-sieving-prime-pi1grim2019 > sh > ≡ sieve3_2.o
1 func=sieve3, pnum=64, count=455052511, time=0.455581
≡ sieve3_3.o × starter.py
cs211-hw3-sieving-prime-pi1grim2019 > sh > ≡ sieve3_3.o
1 func=sieve3, pnum=96, count=455052511, time=0.304960
≡ sieve3_4.o × starter.py
cs211-hw3-sieving-prime-pi1grim2019 > sh > ≡ sieve3_4.o
1 func=sieve3, pnum=128, count=455052511, time=0.228466
≡ sieve3_5.o × starter.py
cs211-hw3-sieving-prime-pi1grim2019 > sh > ≡ sieve3_5.o
1 func=sieve3, pnum=160, count=455052511, time=0.183457

```

The execution time is inversely proportional to the number of cores.



Sieve3 has achieved approximately a 13x speedup compared to Sieve0.